



## Research article

# Entropy space quantum behaved dung beetle optimized long short-term memory network for industrial process fault diagnosis

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## ABSTRACT

To address the challenges posed by the high dimensionality and strong nonlinearity of complex industrial process data, as well as the inherent defects in feature learning and hyperparameter dependency of the traditional long short-term memory network (LSTM), this paper proposes a novel hierarchical fault diagnosis method termed entropy space quantum behaved dung beetle optimized LSTM (ES-QLSTM). The proposed method employs kernel entropy component analysis (KECA) to extract features from raw data in the entropy space (ES). Benefiting from the strong nonlinear mapping capability and high-order information retention property of the KECA, the proposed method effectively preserves critical high-order statistical information that cannot be captured by conventional methods, which compensates for the insufficient feature mining ability of the LSTM. Then, this proposed method integrates entropy space feature extraction with a quantum behaved mechanism into the dung beetle optimization algorithm, constructing a systematic synergistic framework for entropy space feature dimensionality reduction and intelligent hyperparameter optimization. This framework solves two key shortcomings of the LSTM simultaneously, weak feature extraction from high-dimensional nonlinear data and unstable performance caused by empirical hyperparameter settings. Subsequently, a quantum behaved mechanism is incorporated into the dung beetle optimization (DBO) algorithm, resulting in the quantum behaved dung beetle optimization (QDBO) algorithm, which is utilized to optimize the key hyperparameters of the LSTM network. The ultimately constructed ES-QLSTM method enhances diagnostic performance through the synergistic integration of entropy-space feature purification and intelligent hyperparameter optimization. Experimental validation on the Tennessee Eastman Process (TEP) and grid-connected photovoltaic system (GPVS) datasets demonstrates that the proposed method outperforms traditional models and improved models in diagnosing complex faults, with the average diagnostic accuracy reaching 93.70% and 83.91% on the two datasets, respectively.

## 1. Introduction

The reliable and safe operation of modern industrial systems is largely contingent upon the efficacy of fault diagnosis technologies [1]. With the in-depth advancement of Industry 4.0 and smart manufacturing, the volume of data generated in industrial production processes has experienced explosive growth, which is typically featured by complex attributes such as high dimensionality, nonlinearity, and

dynamic coupling [2]. Conventional fault diagnosis approaches are highly dependent on the establishment of accurate mechanistic models, yet their inadequate adaptive capacity makes it hard to capture fault-related features accurately when confronted with the complex data, thus restricting their diagnostic performance [3]. To address these issues, this paper integrates entropy space feature reduction and quantum behaved optimization, aiming to provide a reliable solution for real-time industrial fault diagnosis.

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Given the prevalent nonlinearity of the industrial data, traditional linear dimensionality reduction methods were inadequate in fully revealing the inherent correlations within the data. Schölkopf et al. [4] proposed the kernel PCA (KPCA) for mapping the data to a high-dimensional feature space through a nonlinear transformation and then performed linear dimensionality reduction in this space, which effectively overcame the modeling constraints of linear methods and enhanced the adaptability of feature extraction to nonlinear data. Fezai et al. [5] further proposed an online KPCA algorithm that simplified kernel matrix computations to enable real-time monitoring, thereby resolving the problem that the traditional KPCA was not suitable for industrial online application scenarios. Although the KPCA outperformed the PCA in nonlinear modeling, its principal component selection mechanism relied solely on eigenvalue magnitude. This often led to the neglect of higher-order statistical features, resulting in the loss of critical fault information. To mitigate this limitation, Jenssen [6] pioneered the kernel entropy component analysis (KECA) framework, which employed Rényi entropy as the metric for feature selection, which overcame the over-reliance of the KPCA on eigenvalues and ensured the preservation of higher-order information in the data. Li et al. [7] combined the KECA with a moving window to achieved adaptive fault detection during dynamic processes. Despite these improvements, the potential of entropy-space features to guide the hyperparameter search landscape of downstream classifiers remains under-explored.

In the construction of fault classification models, long short-term memory (LSTM) networks leverage their unique gating mechanisms to effectively capture long-term dependencies in time-series data and mitigate the vanishing gradient problem inherent in traditional recurrent neural networks [8]. Consequently, the LSTM is widely applied for fault pattern learning in dynamic industrial data [9]. The LSTM is adopted for its unique gating mechanism that captures long-term temporal dependencies and mitigates gradient vanishing, outperforming traditional methods in handling dynamic industrial time-series data. However, the LSTM performance is highly dependent on hyperparameter tuning, such as learning rate, number of hidden layer neurons, and L2 regularization coefficient. Traditional manual tuning or grid search methods are prone to getting stuck in local optima due to limited search ranges, resulting in insufficient model generalization capabilities [10]. Therefore, Liu et al. [11] employed a particle swarm optimization (PSO) algorithm to optimize the LSTM network and combined it with an attention mechanism to enhance the accuracy of new energy cable fault prediction. Gao et al. [12] employed an improved beluga whale optimization algorithm to optimize the LSTM, enhancing the balance factor through nonlinear functions for ship diesel engine piston ring fault diagnosis. Wang et al. [13] utilized an improved sparrow search algorithm to optimize the LSTM, incorporating Tent chaotic mapping and dynamic stride to improve the accuracy and stability of fault data recovery for dry-type transformer temperature sensors. Yousaf et al. [14] employed Bayesian optimization to tune LSTM hyperparameters, combined with a discrete wavelet transform for feature extraction, further improving fault recognition accuracy. Y et al. [15] proposed a multi-information fusion method based on attention collaborative stacked LSTM and quantum particle swarm optimization algorithms to address the challenge of extracting fault features from neutral-point-clamped three-level inverters under non-stationary conditions. Zeng et al. [16] proposed the dual attention LSTM integrating the autoencoder encoding-decoding architecture and dual attention modules to achieve efficient learning of dynamic features and key information in long-term industrial process data. Wang et al. [17,18] proposed an innovative filtering method for lithium-ion battery state of power evaluation and an improved parameter identification strategy for state-of-charge estimation of lithium-ion batteries in electric vehicles, and the full-parameter online identification logic included therein had further promoted the research on mathematical analysis and parameter optimization. To further balance the global exploration and local exploitation capabilities of optimization algorithms, Xue et al. [19]

inspired by the natural behaviors of dung beetles, such as ball rolling, dancing, reproduction and stealing, proposed the dung beetle optimization (DBO) algorithm, which combined the differential evolution mechanism with the collaboration strategy of the biological group, and exhibited a good diversity preservation and convergence performance in multimodal function optimization. However, in the hyperparameter search scenario of industrial fault diagnosis models, the classical DBO suffers from the problems of easy to fall into local extremes in the later iterations and insufficient adaptability to the complex search space [20]. To address the issue of population diversity degradation in traditional swarm intelligence algorithms, Wang et al. [21] proposed an adaptive small-family population-guided swarm intelligence optimization algorithm, which dynamically adjusted the population structure and introduces family-based guidance strategies to enhance global search capability and convergence accuracy. Meanwhile, the quantum computing technology shows potential in breaking through the performance bottleneck of traditional optimization algorithms due to its parallel search and quantum tunneling characteristics. Agrawal et al. proposed the quantum adaptive mutation operator PSO, which extended the search region by combining quantum bit dynamics and adaptive mutation, and performed well on the IEEE-CEC–2022 benchmark problem [22].

In summary, while existing research has made progress in feature extraction and model optimization for industrial fault diagnosis, the nonlinear and redundant nature of high-dimensional industrial data often leads to insufficient feature learning of the LSTM. Meanwhile, hyperparameter optimization algorithms for fault classification models are prone to unstable convergence, failing to provide optimal parameter configurations and thereby compromising diagnostic performance. Therefore, identifying that existing research lacks a unified framework that balances high-order information preservation with quantum-probabilistic global exploration, this paper proposes a novel systematic fault diagnosis paradigm named entropy space quantum behaved dung beetle optimized LSTM (ES-QLSTM), which is a theoretically grounded framework designed to overcome the inherent weaknesses of the LSTM. The proposed method first employs KECA to extract primary features from industrial process data within an entropy space, retaining high-order information and providing high-quality inputs to make up for the lack of nonlinear feature learning in the LSTM. Then, a quantum behavior mechanism is introduced into the traditional dung beetle optimization algorithm, resulting in the QDBO algorithm, which optimizes the key hyperparameters of the LSTM to eliminate performance fluctuations caused by manual parameter tuning. Finally, the ES-QLSTM method synergistically enhances the fault diagnosis performance of industrial processes by deeply integrating entropy space feature extraction with quantum behavior optimization mechanisms, achieving efficient and stable fault diagnosis. The primary contributions of the paper are elaborated in detail as follows.

- 1) A novel entropy space feature extraction strategy based on the KECA is proposed by taking Rényi entropy as the core evaluation metric. While eliminating redundant and irrelevant variables to reduce data complexity, it effectively preserves the key information entropy features closely related to industrial fault patterns, laying a solid foundation for accurate fault identification.

- 2) An improved QDBO algorithm is developed by integrating a quantum behavior mechanism into the original DBO algorithm. The introduced quantum behavior enables particles to achieve quantum hopping in the search space, significantly expanding the exploration range and avoiding the defect of traditional DBO algorithms that are prone to falling into local optima. This improvement enhances the efficiency and accuracy of the algorithm search, which has been verified on multiple benchmark functions.

- 3) Corresponding experimental validations are conducted on two representative industrial datasets including Tennessee Eastman Process (TEP) and grid-connected photovoltaic system (GPVS). The results demonstrate that the proposed ES-QLSTM model outperforms existing

**Table 1**

Performance comparison of the proposed model using different kernel functions.

| Kernel Function | TEP Accuracy (%) | GPVS Accuracy (%) |
|-----------------|------------------|-------------------|
| Gaussian        | 93.70            | 83.91             |
| Polynomial      | 78.42            | 69.35             |
| Sigmoid         | 75.18            | 65.72             |

methods in fault identification accuracy, verifying its promising application prospects in practical industrial fault diagnosis.

The structure of the rest of this article is organized as follows. Section II describes the overall framework and implementation details of the ES-QLSTM fault diagnosis methodology. Section III conducts a comparative analysis of multiple optimization algorithms using four benchmark functions. Section IV demonstrates the experimental results and analyses on the TEP and GPVS datasets. and Section V summarizes the findings of this article and looks forward to the future research directions.

## 2. The proposed method

This segment offers a detailed elaboration on two core components. One is the fault diagnosis based on entropy space feature extraction, and the other is the QDBO Algorithm. Following this, we lay out the full operational workflow of the fault diagnosis scheme that relies on the ES-QLSTM architecture.

### 2.1. Fault diagnosis based on entropy space feature extraction

As a nonlinear data processing method, KECA achieves transformation and dimensionality reduction by projecting data into the kernel feature space and then onto the KPCA principal axes.

We define  $X : (x_1, x_2, \dots, x_n)$  as an  $d \times n$  matrix, with  $n$  representing the sample size and  $d$  indicating the dimension of damage features. For probability density distribution  $g(x)$ , the corresponding Rényi entropy is governed by the equation:

$$H(q) = -\log \int q^2(x) dx \quad (1)$$

This form corresponds to the quadratic Rényi entropy, which is widely adopted in industrial fault diagnosis due to its high sensitivity to subtle changes in fault-related features. Unlike traditional variance-based metrics, quadratic Rényi entropy quantifies the information richness of high-dimensional nonlinear data by integrating the squared probability density, making it more capable of capturing weak coupling fault information that is easily lost in conventional dimensionality reduction.

Due to the monotonicity of the logarithmic function, the calculation of Rényi entropy can be simplified by analyzing. This formula can alternatively be restated as  $V(g) = \varepsilon_g(g)$ , where  $\varepsilon_g(\cdot)$  stands for the expectation operation on the probability density function  $g(x)$ . To accurately estimate  $V(g)$  and  $H(g)$  for noisy industrial data, the Parzen window method is utilized, with the probability density estimation formula given by:

$$\hat{q}(x) = \frac{1}{n} \sum_{x_t \in X} k_c(x, x_t) \quad (2)$$

Here,  $k_c(x, x_t)$  denotes Gaussian kernel function with  $c$  as its tuning parameter. The Gaussian kernel is selected for its infinite differentiability and local similarity preservation, which can effectively model the nonlinear relationships between industrial process variables while suppressing sensor noise. To quantitatively justify the selection of the Gaussian kernel, we evaluated the diagnostic performance of the ES-QLSTM using different kernel functions on both TEP and GPVS datasets. As shown in Table 1, the Gaussian kernel consistently outperforms others.

An estimate of  $\hat{V}(g)$  is then derived by approximating the expectation operator with a sample mean.

$$\hat{V}(q) = \frac{1}{n} \sum_{x_t \in X} \hat{q}(x_t) = \frac{1}{n} \sum_{x_t \in X} \frac{1}{n} \sum_{x'_t \in X} k_c(x_t, x'_t) = \frac{1}{n^2} I^T K I \quad (3)$$

This formula reveals the direct link between Rényi entropy estimation and the kernel matrix  $K$ : the double summation term constructs an  $n \times n$  kernel matrix  $K$ , and  $\hat{V}(g)$  is proportional to the quadratic form  $I^T K I$  of the all-ones vector  $I$ . This lays the mathematical foundation for the KECA to extract features based on entropy contribution, rather than relying solely on eigenvalue magnitude like the KPCA.

The elements of the  $n \times n$  kernel matrix  $K$  are  $k_c(x_t, x'_t)$ , where  $I$  is an  $n \times 1$  all-ones vector. Therefore, using only the available samples present in the corresponding elements of kernel matrix  $K$ , the estimated Rényi entropy can be evaluated. Furthermore, the estimate of Rényi entropy can be expressed in terms of the eigenvalues and eigenvectors of the kernel matrix, which can undergo eigenvalue decomposition as shown in Eq. (4).

Furthermore, the Rényi entropy estimate can be expressed using the eigenvalues and eigenvectors of the kernel matrix, which undergoes eigenvalue decomposition as shown below:

$$K = \Lambda E \Lambda^T \quad (4)$$

where  $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$  is the eigenvalue matrix and  $E = (e_1, \dots, e_n)$  is the eigenvector matrix. Based on this decomposition, the Rényi entropy estimate and entropy contribution metric  $\psi_i$  are respectively given by:

$$\hat{V}(q) = \frac{1}{n^2} \sum_{i=1}^n (\sqrt{\lambda_i} e_i^T I)^2 \quad (5)$$

$$\psi_i = (\sqrt{\lambda_i} e_i^T I)^2 \quad (6)$$

Although Rényi entropy estimation depends on both eigenvalues and eigenvectors, a high eigenvalue is not guaranteed to contribute most to information entropy. Consequently,  $\psi_i$  from Eq. (6) represents the Rényi entropy of individual elements in Eq. (5), and the values generated by Eq. (6) help to compute Rényi entropy estimates.

For the input dataset  $X : (x_1, x_2, \dots, x_n)$ , it is first mapped into the kernel feature space via a nonlinear function  $\phi(\cdot)$ , resulting in the transformed dataset  $\Phi = [\phi(x_1), \phi(x_2), \dots, \phi(x_n)]$  in the kernel space. The principal axes constructed in KECA — which maximize the retention of Rényi entropy estimates — form a dominant subspace carrying most of the original data information. By projecting  $\Phi$  onto this dominant subspace  $U_k$ , the  $n$ -dimensional data is transformed into low-dimensional features, as shown in Eq. (7):

$$T_k = U_k \Phi = \Lambda_k^{\frac{1}{2}} E_k^T \min_{\lambda_1, e_1, \dots, \lambda_n, e_n} \hat{V}(q) \sim \hat{V}_k(q) \quad (7)$$

where  $T_k = [t_1, t_2, t_3, \dots, t_k]$  is the score matrix,  $\Lambda_k = \text{diag}(\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_k)$  is the eigenvalue matrix, and  $E_k = [e_1, e_2, e_3, \dots, e_k]$  is the eigenvector matrix. Through the above steps, the KECA extracts the components contributing most significantly to data entropy within the high-dimensional kernel feature space. This achieves nonlinear dimensionality reduction while effectively preserving critical data information, providing low-dimensional, efficient feature inputs for subsequent fault diagnosis model construction.

Features reduced by the KECA retain significant temporal correlations. LSTM networks, owing to their unique memory cell structure, can simultaneously capture both long-term and short-term dependencies among process variables. This effectively mitigates gradient vanishing and gradient explosion issues, making LSTMs more suitable for diagnosing industrial process failures characterized by strong temporal correlations and dynamic coupling properties.

The LSTM network requires input data to be in a specific sequence format. Therefore, after completing feature extraction through the

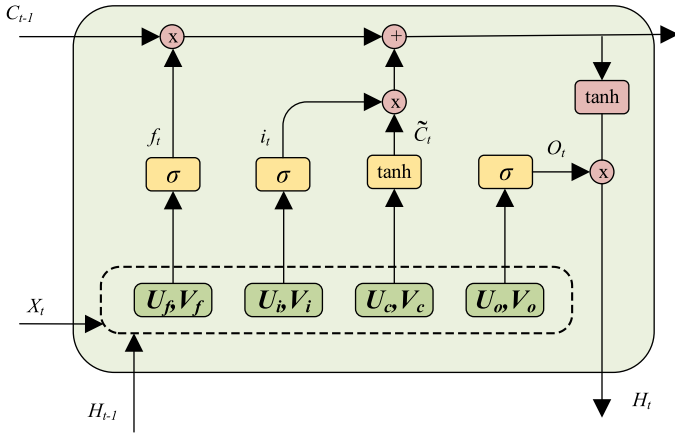


Fig. 1. Schematic diagram of LSTM.

KECA, it is necessary to preprocess the low-dimensional feature matrix  $T_k$  to adapt to the input requirements of the LSTM. The specific steps are shown as follows:

$$P_{train} = T_k^T = [p_1, p_2, \dots, p_n] \in \mathbb{R}^{k \times n} \quad (8)$$

each column  $P_{train}(:, i) \in \mathbb{R}^{k \times 1}$  represents the  $t$ -dimensional feature vector of the  $i^{\text{th}}$  sample.

Use min-max normalization to map features to the range 0–1:

$$\tilde{P}_{train} = \text{map min max}(P_{train}, 0, 1) \quad (9)$$

Convert the normalized feature matrix into the sequence format of cell arrays required by the LSTM. For each sample  $p_i \in \mathbb{R}^{k \times 1}$ , organize it into a sequence of length  $k$ , where the feature dimension at each time step is 1:

$$P_{train} = \{p_1, p_2, \dots, p_n\} \quad (10)$$

Each  $p_i$  represents a sequence unit:

$$p_i = [x_1^{(i)}, x_2^{(i)}, \dots, x_k^{(i)}]^T \in \mathbb{R}^{k \times 1} \quad (11)$$

The  $k$  elements in  $p_i$  correspond to  $k$  time steps. with each time step having a feature dimension of 1. This organization enables the LSTM to capture latent correlations and dependencies among the  $k$  reduced-dimension features.

Under this serialization format, the input for the  $i$ -th sample at time step  $t$  is  $x_t^{(i)}$ , the  $t$ -th dimensionality-reduced feature of that sample. The LSTM processes the sequence step-by-step through the gating mechanisms defined by Eqs. (12)–(17), capturing dependencies among  $k$  features. It ultimately outputs the hidden state  $H_k$  at the final time step, which serves as the feature representation for that sample in the final classification task. Under this serialization format, the input for the  $i$ -th sample at time step  $t$  is  $x_t^{(i)}$ , the  $t$ -th dimensionality-reduced feature of

that sample. The LSTM processes the sequence step-by-step through the gating mechanisms defined by Eqs. (12)–(17), capturing dependencies among  $k$  features. It ultimately outputs the hidden state  $H_k$  at the final time step, which serves as the feature representation for that sample in the final classification task. As shown in Fig. 1, the LSTM realizes temporal feature modeling through three gating mechanisms: the forget gate, the input gate and the output gate. They control information forgetting, updating and output respectively to mitigate gradient vanishing. The key formulas and parameter explanations are as follows:

The forget gate is shown as follows:

$$f_t = S(U_f X_t + V_f H_{t-1} + \delta_f) \quad (12)$$

where  $f_t$  is the forget gate output,  $S(\cdot)$  denotes the sigmoid function,  $U_f, V_f$  are coefficient matrices,  $\delta_f$  is the bias vector,  $X_t$  is the current input, and  $H_{t-1}$  is the previous hidden state. A value of 1 means full information retention, while 0 means complete forgetting.

The input gate is shown as follows:

$$i_t = S(U_i X_t + V_i H_{t-1} + \delta_i) \quad (13)$$

$$\tilde{C}_t = \tanh(U_c X_t + V_c H_{t-1} + \delta_c) \quad (14)$$

where  $i_t$  is the update coefficient,  $\tilde{C}_t$  is the candidate cell state, and  $\tanh(\cdot)$  denotes the hyperbolic tangent function.

The output gate is shown as follows:

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (15)$$

$$O_t = S(W_o X_t + U_o H_{t-1} + \delta_o) \quad (16)$$

$$H_t = O_t \odot \tanh(C_t) \quad (17)$$

where  $O_t$  is the output coefficient,  $C_t$  is the updated cell state,  $\odot$  denotes element-wise multiplication, and  $H_t$  is the current hidden state.

The LSTM network processes the sequence input  $X_t$  step by step through the above gating mechanism, and the output  $H_t$  of the final time step is used as the high-level feature representation of the sample, which is provided to the subsequent fully connected layer for fault classification. However, the performance of the LSTM is highly dependent on hyperparameters such as initial learning rate, number of hidden layer nodes, and L2 regularization coefficient. Therefore, it is necessary to optimize these hyperparameters to improve the model generalization ability and fault diagnosis accuracy.

## 2.2. LSTM parameter optimization based on QDBO algorithm

The initial learning rate, number of hidden layer nodes, and L2 regularization coefficient are selected as optimization objects, as they are widely confirmed to be the most performance-sensitive hyperparameters for the LSTM in industrial fault diagnosis [23], directly governing model convergence, feature learning capacity, and generalization. In contrast, dropout rate, number of layers, and optimizer type

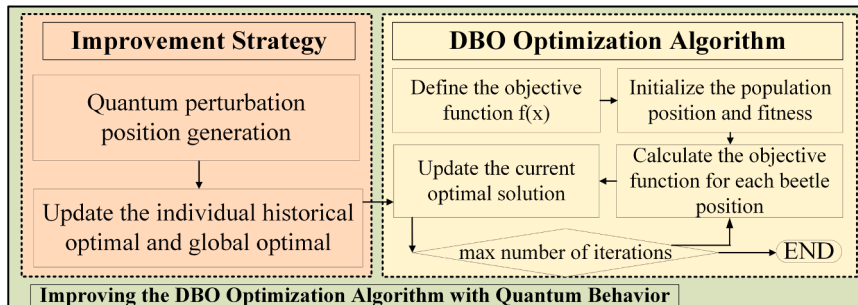


Fig. 2. DBO Improvement Schematic Diagram.

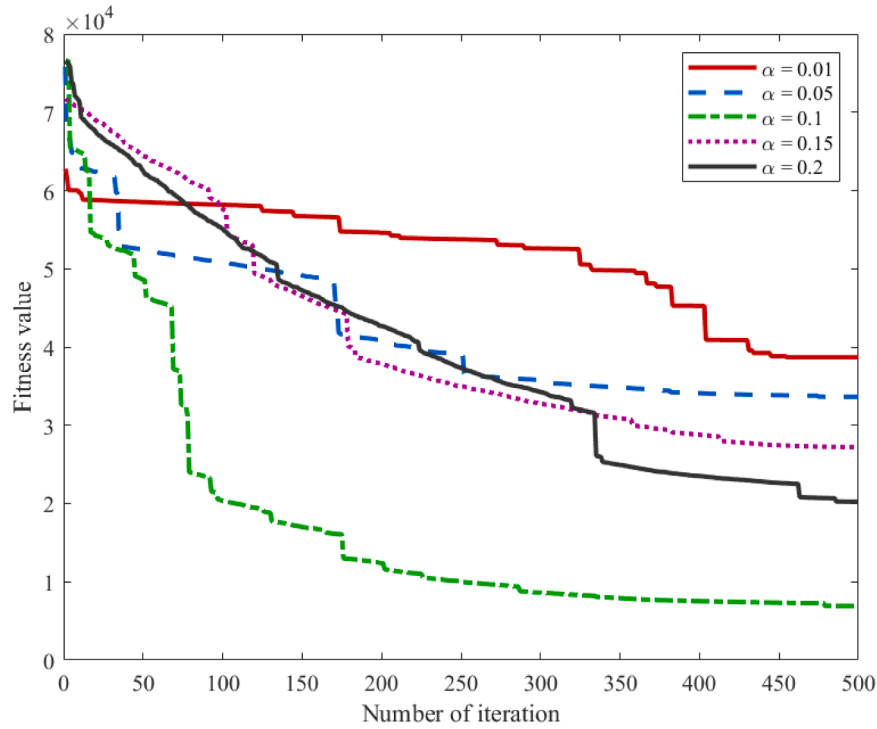


Fig. 3. Sensitivity Analysis of the Step-Size Coefficient  $\alpha$ .

have negligible impacts on industrial time-series data due to its weak noise and inherent temporal dependencies, making additional optimization unnecessary. For the QDBO algorithm, the selection of 20% ball-rolling beetles and 20% quantum perturbation proportion follows the population configuration of the original DBO algorithm [24], which balances local exploitation and global exploration without arbitrary parameter setting. To address the issue of LSTM model hyperparameters

being empirically set, which often leads to poor generalization capabilities, this article introduces a quantum behavior mechanism based on the traditional dune beetle optimization algorithm, proposing the quantum behavior-enhanced dune beetle optimization. This algorithm retains DBO advantage of achieving global optimization by simulating dung beetle behaviors such as rolling balls, foraging, and reproduction. Simultaneously, quantum behaviors enhance search randomness and

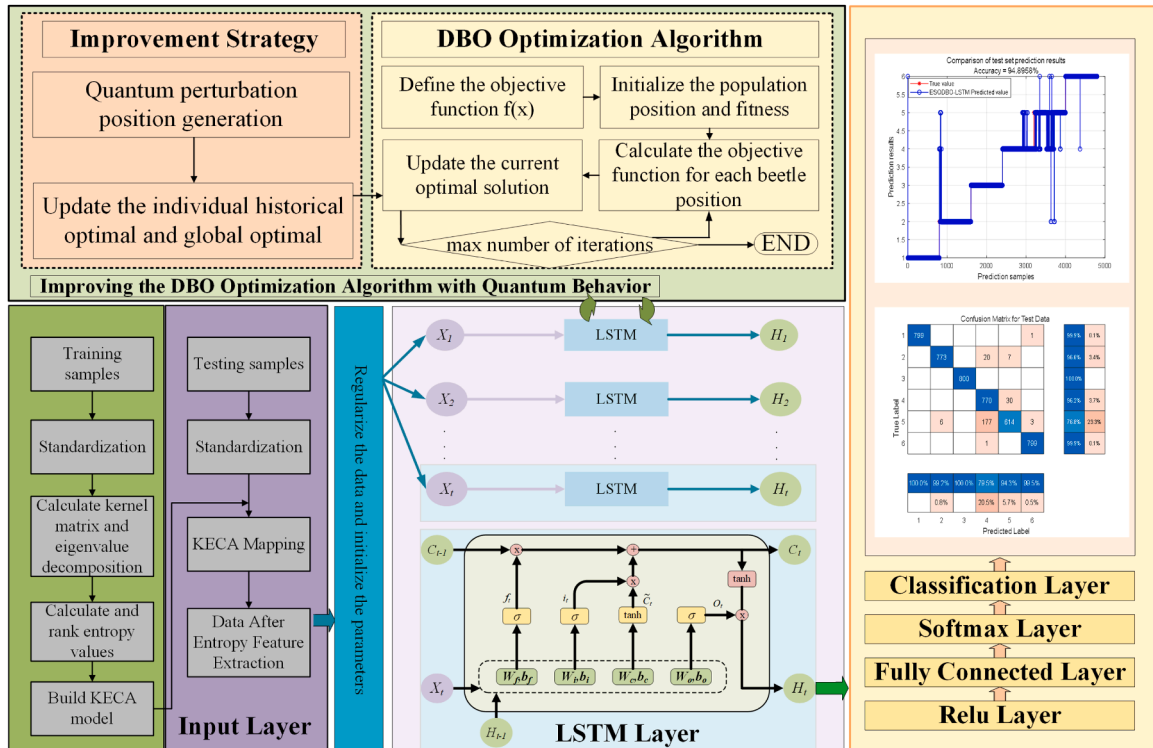


Fig. 4. The ES-QLSTM based Fault Diagnosis Structure.



**Table 2**

Summary of the CEC2017 test functions.

| No. | Benchmark Functions                      | $F_i = F_i(X^*)$ | Dim | Search Interval |
|-----|------------------------------------------|------------------|-----|-----------------|
| 1   | Shifted and Rotated Bent Cigar Function  | 100              | 30  | [−100,100]      |
| 5   | Shifted and Rotated Rastrigin's Function | 500              | 30  | [−100,100]      |
| 13  | Hybrid Function 3 (N = 3)                | 1300             | 30  | [−100,100]      |
| 28  | Composition Function 8 (N = 6)           | 2800             | 30  | [−100,100]      |

global exploration capabilities, effectively addressing the core shortcomings of the traditional DBO. a simplistic individual update mechanism leading to slowed convergence in later stages, limited beetle behaviors prone to local optima, and reduced population diversity during iterations, resulting in insufficient exploration. Therefore, the following improvement strategies is illustrated in Fig. 2.

Define the LSTM hyperparameter optimization variables as a three-dimensional vector.

$$\theta = [\alpha, h, \omega]^T \quad (18)$$

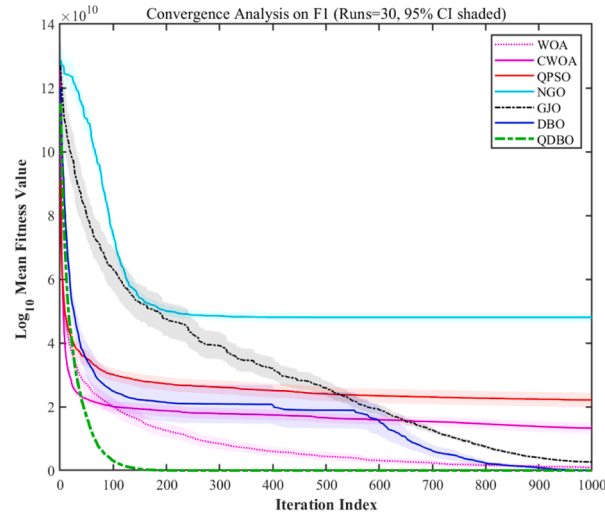
where  $\alpha \in [\alpha_{min}, \alpha_{max}]$  is the initial learning rate,  $h \in [h_{min}, h_{max}]$  is the number of nodes in the LSTM hidden layer, and  $\omega \in [\omega_{min}, \omega_{max}]$  is the L2 regularization coefficient.

The classification error rate of the LSTM model on the training set serves as the fitness function for the QDBO algorithm, measuring the quality of hyperparameter combinations. For a given hyperparameter combination  $\theta$ , construct the LSTM network  $f(\cdot; \theta)$  and evaluate it on the training set:

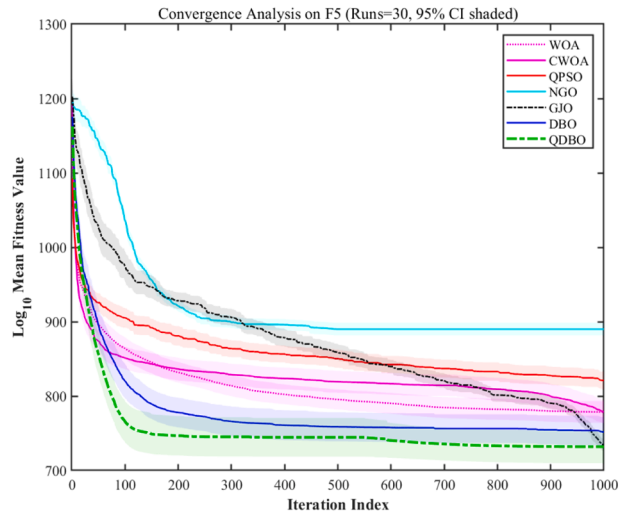
$$F(\theta) = \frac{1}{n} \sum_{i=1}^n I(\hat{c}_i \neq c_i) \times 100\% \quad (19)$$

where  $c_i = f(p_i; \theta)$  denotes the model-predicted category,  $c_i$  represents the true category, and  $I(\cdot)$  is the indicator function.

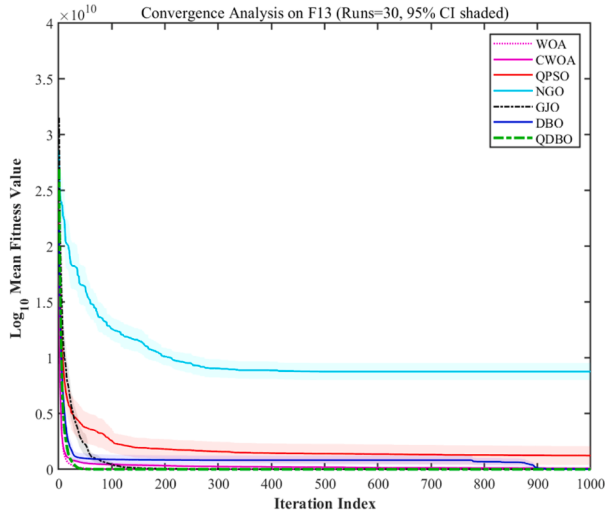
The QDBO algorithm achieves basic optimization by simulating the dung beetle ball-rolling and foraging behaviors, while incorporating quantum behavior to enhance global exploration capabilities. Randomly generate  $N$  candidate hyperparameter solutions covering the entire



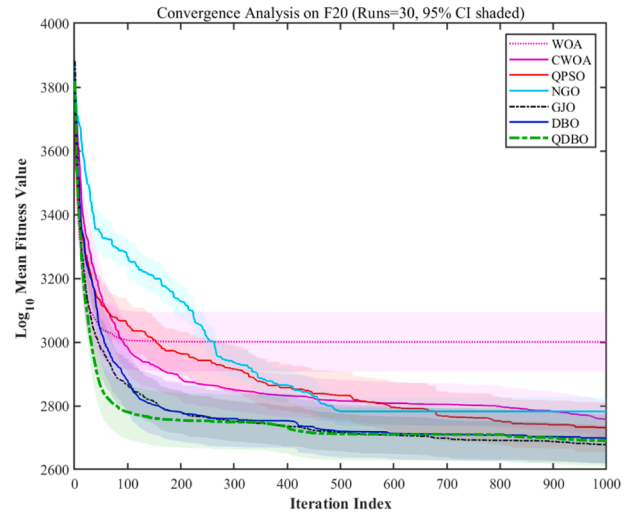
(a)



(b)



(c)



(d)

**Fig. 5.** Performance comparison of distinct algorithms on CEC2017 test functions: (a) F1 (single-modal test function), (b) F5 (basic multi-modal test function), (c) F13 (hybrid test function), (d) F28 (composite test function).

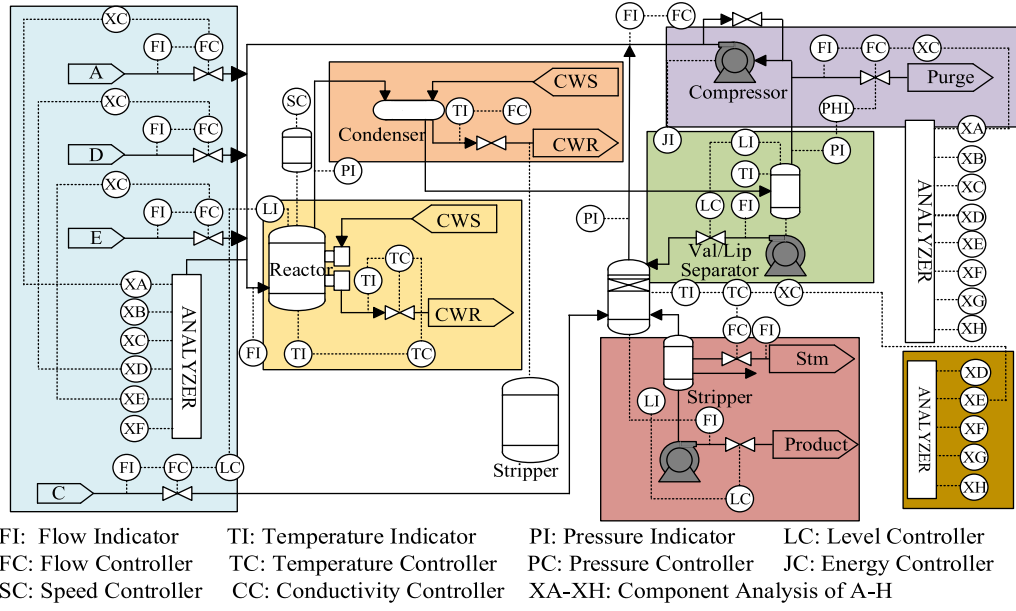


Fig. 6. TEP flow chart.

Table 3

Fault categories and definitions in the tennessee eastman process.

| Fault ID | Description                                           | Fault Type |
|----------|-------------------------------------------------------|------------|
| IDV 4    | Reactor cooling water inlet temperature deviation     | Step       |
| IDV 5    | Condenser cooling water inlet temperature fluctuation | Step       |
| IDV 7    | Pressure loss in header C                             | Step       |
| IDV 10   | Temperature anomaly in feed stream C                  | RV         |
| IDV 12   | Condenser cooling water inlet temperature drift       | RV         |
| IDV 14   | Sticking fault in reactor cooling water control valve | Sticking   |

search space:

$$\theta_i^{(0)} = lb + (ub - lb) \cdot r_i, \quad i = 1, 2, \dots, N \quad (20)$$

where  $r_i$  is a 0–1 uniform random vector,  $lb = [\alpha_{\min}, h_{\min}, \omega_{\min}]^T$  and  $ub = [\alpha_{\max}, h_{\max}, \omega_{\max}]^T$  respectively are the hyperparameter lower and upper bound vectors.

The top 20% of individuals in the population was selected as ball-rolling dung beetles. Their positions were updated by simulating their behaviors to improve the local exploitation of the algorithm.

$$\theta_{best}^{(t+1)} = \arg\min_i F_i^{(t)} \quad (21)$$

$$\theta_i^{(t)} = \theta_i^{(t-1)} + b \cdot |\theta_i^{(t-1)} - \theta_{worst}^{(t-1)}| + a \cdot 0.1 \cdot \theta_i^{(t-1)} \quad (22)$$

where  $a \in \{-1, 1\}$  is the random direction coefficient,  $b \in [0, 1]$  is the random step size coefficient, and  $\theta_{worst}^{(t-1)}$  denotes the worst solution from generation  $t-1$ , guiding producers toward more favorable regions.

The remaining individuals simulate foraging behavior to converge

Table 4

Test accuracy for fault diagnosis on TEP.

| Fault(%)       | IDV 4 | IDV 5 | IDV 7 | IDV 10 | IDV 12 | IDV 14 | Average |
|----------------|-------|-------|-------|--------|--------|--------|---------|
| CNN            | 67.75 | 52.00 | 91.25 | 47.00  | 85.38  | 99.00  | 73.73   |
| LSTM           | 100   | 91.00 | 100   | 59.50  | 65.13  | 45.75  | 76.89   |
| ES-LSTM        | 98.75 | 67.13 | 98.63 | 65.88  | 71.00  | 99.88  | 83.54   |
| KECA-GS-LSTM   | 99.13 | 88.50 | 98.75 | 85.25  | 66.38  | 95.13  | 88.86   |
| DBO-LSTM       | 99.88 | 94.25 | 100   | 92.38  | 69.88  | 85.25  | 90.27   |
| KPCA-QDBO-LSTM | 99.38 | 93.25 | 99.13 | 91.50  | 63.75  | 99.50  | 91.09   |
| QDBO-LSTM      | 99.88 | 94.50 | 99.88 | 98.25  | 64.63  | 99.88  | 92.83   |
| BDNPEOLPP      | 99.25 | 97.25 | 100   | 97.13  | 36.63  | 96.13  | 87.73   |
| ES-QLSTM       | 99.88 | 97.63 | 99.25 | 97.50  | 68.00  | 100    | 93.70   |

toward the optimal solution, while introducing random perturbations to prevent local convergence:

$$\theta_i^{(t)} = \theta_{best}^{(t)} + r_1 \cdot (\theta_i^{(t-1)} - X_{new1}) + r_2 \cdot (\theta_i^{(t-1)} - X_{new2}) \quad (23)$$

among these,  $\theta_{best}^{(t)}$  denotes the optimal solution of generation  $t$ ,  $r_1$  and  $r_2$  are random vectors, and  $X_{new1}, X_{new2}$  represent randomly selected positions of population individuals.

Quantum behavioral perturbations are introduced to 20% of randomly selected individuals to enhance global exploration capabilities:

$$\theta_q^{(t)} = \theta_{best}^{(t)} + \text{sign}(r - 0.5) \cdot L \cdot \log(1/u) \quad (24)$$

where  $L = 0.1 \cdot (1 - t/T_{\max})$  denotes the feature length that decays with iterations, balancing exploration in the early phase and convergence in the late phase. The attenuation coefficient 0.1 of the feature length  $L$  is selected based on comprehensive performance evaluation on CEC2017 benchmark functions. As shown in Fig. 3, different values of the step-size coefficient lead to distinct convergence behaviors. The value 0.1 achieves the optimal balance between fast convergence and stable optimization performance, avoiding insufficient perturbation from smaller coefficients and excessive randomness from larger coefficients. It avoids insufficient perturbation from smaller coefficients and excessive randomness from larger coefficients ensuring stable optimization for LSTM hyperparameter tuning.  $u, r \in [0, 1]$  represent random vectors, and  $\text{sign}(\cdot)$  denotes the sign function.

Ensure that the updated hyperparameters do not exceed the search

space:

$$\theta_i^{(t)} = \max(\min(\theta_i^{(t)}, ub), lb) \quad (25)$$

Compute the fitness of each individual  $F_i^{(t)} = F(\theta_i^{(t)})$  and update the global optimal solution:

$$\theta_{best}^{(t+1)} = \operatorname{argmin}_i F_i^{(t)} \quad (26)$$

When the iteration count reaches the maximum iteration count  $T_{\max}$ , output the globally optimal hyperparameter combination  $\theta^* = \theta_{best}^{(T_{\max})}$  as the final hyperparameters for the LSTM model. Algorithm 1 constitutes the complete pseudo-code of the QDBO algorithm.

**Algorithm 1.** Algorithm1 Algorithm of the QDBO.

### 2.3. The ES-QLSTM fault diagnosis steps

Addressing the characteristics of high-dimensional nonlinearity and temporal dynamic coupling in industrial process data, as well as issues such as insufficient feature extraction and inefficient parameter optimization in traditional fault diagnosis models, an ES-QLSTM based fault diagnosis method is proposed. The framework of this proposed method is illustrated in Fig. 4. The specific steps of the ES-QLSTM based fault diagnosis are as follows.

**Step 1: Data Preparation:** Collect raw industrial process data, partition it into training and test sets, and separate features from fault labels. Standardize the feature matrix using the z-score method.

**Step 2: Entropy Feature Extraction:** Using standardized training set features as input, construct a kernel matrix between samples with a

---

**Input:** Population size  $N$ , maximum iterations  $Max\_iter$ , search range  $[lb, ub]$ , dimension  $dim$ , fitness function  $fobj$

**Output:** Optimal solution  $bestX$ , optimal fitness  $fMin$ , convergence curve

Initialize population  $x_i$  randomly within  $[lb, ub]$

Calculate fitness  $fit_i = fobj(x_i)$  for each individual

Record historical optimal position  $pX$  and previous generation position  $XX$

**while**  $t \leq Max\_iter$  **do**

    Update adaptive factor  $R = 1 - t/Max\_iter$

    Find the worst individual  $worse$  in current population

**Update producers**

**for**  $i = 1:pNum$  **do**

        If random condition is satisfied:

$$x_i = pX_i + b \cdot |pX_i - worse| + a \cdot 0.1 \cdot XX_i$$

        Else:

$$\theta = \operatorname{rand}(0, \pi)$$

$$x_i = pX_i + \tan(\theta) \cdot |pX_i - XX_i|$$

        Boundary restriction

        Update fitness

Find current best  $bestXX$  and generate new search ranges

**Update non-producers**

    Update individuals by current optimal & historical optimal guidance

    Boundary restriction & fitness update

**Quantum perturbation operation**

$$L = 0.1 \cdot R$$

**for**  $i = 1:num\_q$

        Generate new position by quantum behavior around  $bestX$

        Replace if better

**Update historical and global optimum**

$$XX = pX$$

Update  $pX$ ,  $bestX$ ,  $fMin$

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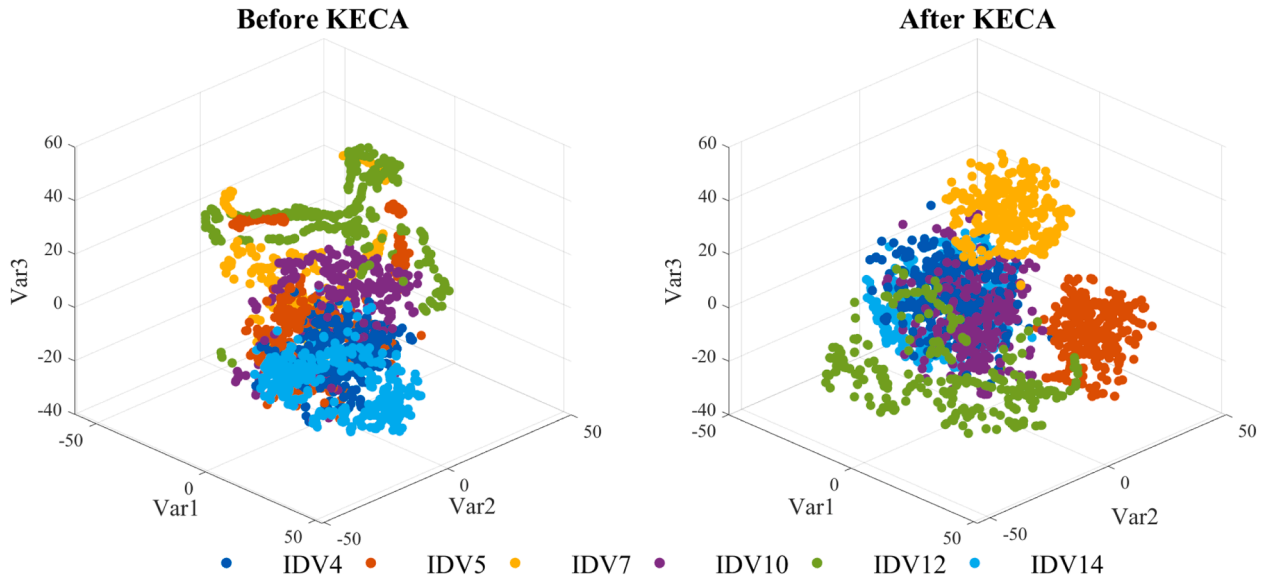


Fig. 7. Visualization graph of the first 3-D features on TEP.

Gaussian kernel function. Apply centering to eliminate the impact of data distribution shifts. Perform eigenvalue decomposition on the centered kernel matrix. Calculate the entropy contribution of each principal component using Rényi entropy theory. Select key principal components in descending order of entropy contribution.

Step 3: Global Optimization of LSTM Hyperparameters Using the QDBO: Configure QDBO algorithm parameters, using the training error of the LSTM on entropy features as the fitness function, and iteratively select optimal hyperparameters that best fit the temporal patterns of entropy features.

Step 4: Training the LSTM Model with Optimal Parameters: Convert the entropy features from the KECA dimensionality reduction into a cell array format compatible with the LSTM. Transform the labels into categorical variables. Combine these with the optimal hyperparameters optimized by the QDBO to construct the LSTM diagnostic model.

Step 5: Fault Classification: Introduce the fused features into the fully connected layer and perform fault classification using the softmax activation function.

### 3. Optimization algorithm comparison

To validate the superiority of the proposed quantum behavior optimization DBO, the CEC2017 function set was employed for testing. This benchmark collection comprises a series of optimization problems of varying types, serving as a standard for evaluating the performance of alternative optimization algorithms. The dataset comprises 30 single-objective test functions categorized into unimodal functions (F1-F3), simple multimodal functions (F4-F10), hybrid functions (F11-F20), and composite functions (F21-F30). For each category, one representative function was selected for comparison as shown in Table 2. Additionally, mainstream optimization algorithms were introduced to contrast with the unmodified DBO.

The results in Fig. 5 demonstrate that on the CEC2017 optimization problems, the quantum-behaved dung beetle optimization (QDBO) algorithm exhibits superior performance compared to a broader set of

advanced hybrid optimization algorithms, including quantum-behaved particle swarm optimization (QPSO), chaotic whale optimization algorithm (CWOA) [25], golden jackal optimization (GJO) [26], northern goshawk optimization (NGO) [27], and the traditional DBO. To ensure statistical reliability, each algorithm is run 30 times independently, and the mean convergence curves with 95% confidence intervals are presented in Fig. 5. As shown in Fig. 5(d), in complex large-scale composite optimization scenarios, the convergence curve of the QDBO algorithm consistently lies below those of all comparative algorithms, including the QPSO, the CWOA, the GJO, the NGO, and the DBO, with significantly faster convergence speed. The QDBO achieves a stable optimal fitness value on the order of  $10^{-1}$  within only about 80 iterations, while the QPSO, the CWOA, the traditional DBO, the GJO, and other algorithms require more than 250 iterations to stabilize, with their final fitness values only reaching the order of  $10^{-3}$ – $10^{-4}$ . Moreover, the shaded confidence intervals for QDBO are the narrowest among all algorithms, indicating its high stability and repeatability across independent runs. Unlike traditional DBO and other advanced hybrid algorithms, which show obvious fluctuations and wider confidence bands, the QDBO exhibits minimal variance after convergence, with the fluctuation amplitude controlled within 5%, fully reflecting its superior global optimization capability and convergence efficiency in complex combinatorial optimization problems.

In summary, the QDBO effectively enhances hyperparameter optimization performance in high-dimensional nonlinear problems by introducing adaptive step sizes and quantum perturbations into the original dung beetle optimization framework. Such approaches not only boost the algorithm global and local search capacities, but also elevate its randomness and population diversity, enabling superior performance in complex environments. Compared with advanced hybrid algorithms, the QDBO achieves faster convergence, lower final fitness values, and better stability, further verifying its effectiveness in the LSTM hyperparameter tuning for industrial fault diagnosis tasks.

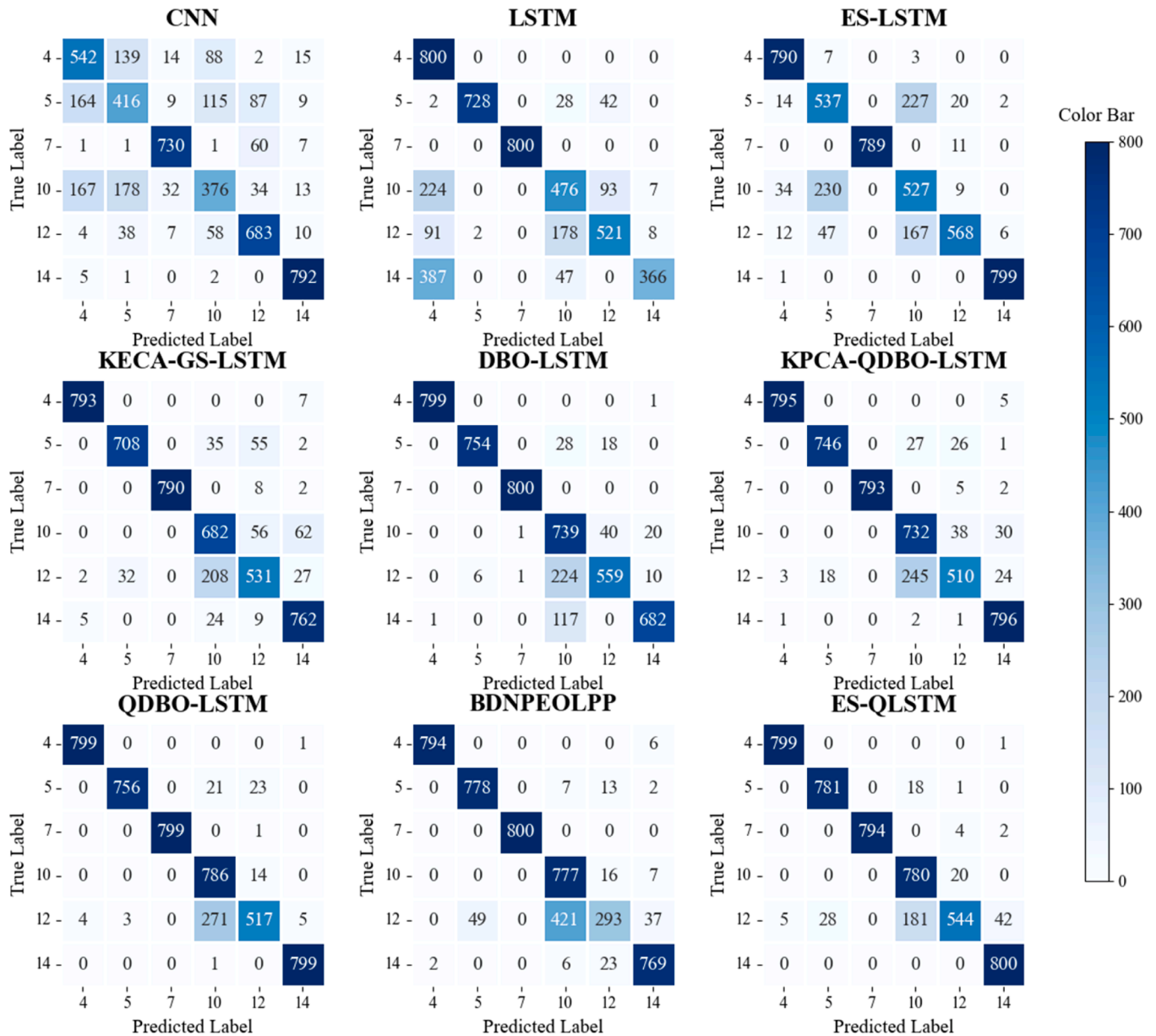


Fig. 8. Fault diagnosis on the TEP.

## 4. Case studies

### 4.1. The TEP

The TEP is a well-established benchmark for complex chemical processes and has long served as a standard testbed in fault diagnosis and process optimization research [28]. As illustrated in Fig. 6, the TEP configuration includes four gaseous feed streams, two liquid product streams, a single byproduct stream, and a conversion gas stream, with its operational behavior described by a total of 52 process variables [29]. The test suite encompasses 21 distinct fault scenarios, grouped into six broad categories, whose detailed characteristics are documented in reference [30]. For each fault category, the corresponding dataset is structured to support model development and validation: 480 samples are allocated for training, and 960 samples are reserved for testing, with process measurements recorded at 3-minute intervals.

To validate the effectiveness of the ES-QLSTM method, six fault types

from Table 3 are selected for testing [31]. In the TEP, 1200 samples are collected for each fault type, with the training and test sets split in a 1:2 ratio.

Table 4 shows that the ES-QLSTM model performs optimally in all six fault scenarios of TEP fault diagnosis, with an average accuracy of 93.70%, which is significantly better than that of the CNN with 73.73%, and the LSTM with 76.89%. Fig. 6 presents the three-dimensional feature vector plot, which reveals significant data overlap between IDV10 and IDV12 in fault diagnosis, primarily due to their strong coupling, as both faults are temperature-related. For the randomly fluctuating faults IDV10 and IDV12, traditional models and single deep learning models exhibit extremely low recognition accuracy. IDV10 achieves 47.00%, and 59.50% accuracy under the CNN, and the LSTM, respectively, whereas the ES-QLSTM achieves 97.50% fault diagnosis accuracy. For IDV12, the LSTM achieves 65.13% accuracy, while the ES-QLSTM reaches 68.00%. the KECA-LSTM averages 83.54%, the DBO-LSTM with 90.27%, and the QDBO-LSTM with 92.83% all lower than

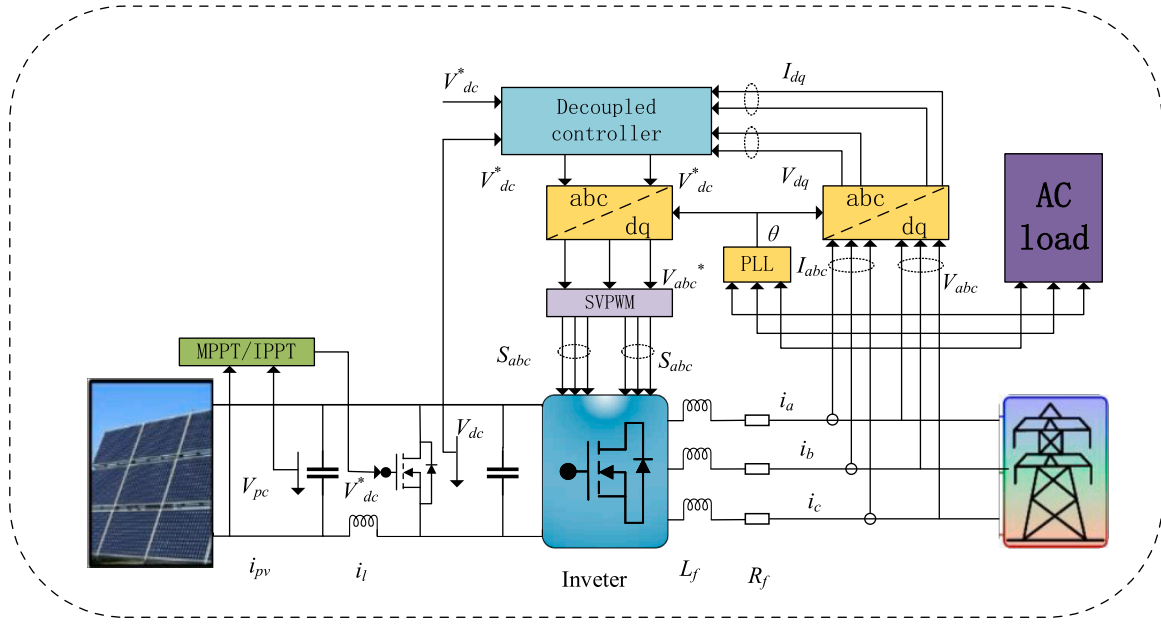


Fig. 9. Schematic diagram of the grid-connected photovoltaic system process flow.

Table 5

Fault category definitions for GPVS experiments.

| Fault ID | Fault Type                              |
|----------|-----------------------------------------|
| F1L      | Inverter module malfunction             |
| F2L      | Feedback sensor signal anomaly          |
| F3L      | Abnormal grid operating conditions      |
| F4L      | Partial mismatch in PV array output     |
| F5L      | PV array performance inconsistency      |
| F6L      | Malfunction in MPPT/IPPT control unit   |
| F7L      | Fault in boost converter control module |

the ES-QLSTM. The temperature coupling characteristics of IDV10 and IDV12 make it difficult for traditional methods to distinguish fault features, while entropy space feature extraction preserves the high-order statistical information of temperature signals based on Rényi entropy and achieves effective decoupling of coupled signals. The low learning rate and adaptive hidden layer node parameters obtained by the QDBO algorithm make the LSTM more sensitive to the temporal patterns of random temperature fluctuations and slow drifts, thus achieving significantly higher diagnostic accuracy for these two faults. The 40.67% accuracy result of F2L from sensor noise, feature overlap with F6L, and insufficient noise suppression and coupled fault differentiation. For IDV5 faults, the KECA-LSTM achieves only 67.13% accuracy, where

the ES-QLSTM reaches 97.63%, fully demonstrating the synergistic advantages of entropy feature extraction and quantum behavior parameter optimization. Additionally, the recent advanced dimensionality reduction-based fault diagnosis method with the BDNPEOLPP [32] achieves an average accuracy of 87.73% on the TEP dataset, which is lower than the ES-QLSTM 93.70%. Moreover, the ES-QLSTM outperforms the BDNPEOLPP significantly in IDV12 68.00% versus with 36.63% and IDV14 100% versus with 96.13% faults, demonstrating stronger robustness to drift and sticking faults.

As shown in the confusion matrix for fault diagnosis in Fig. 8, traditional models such as the CNN, and the LSTM exhibit significant misclassification issues. For random fluctuating faults IDV10 and IDV12, these models demonstrate inadequate classification performance, with a large number of samples incorrectly classified as normal or other fault types. This pattern is clearly visible in the confusion matrix, where the off-diagonal entries for these fault types make up a considerable share of the total. By comparison, the confusion matrix derived from the ES-QLSTM model shows a marked increase in the proportion of diagonal entries across all fault type rows. In particular, the number of misclassified samples for IDV10 and IDV12 is drastically lower. When benchmarked against other state-of-the-art models, the ES-QLSTM confusion matrix demonstrates stronger diagonal concentration, which confirms its superior fault classification accuracy. This allows for more precise differentiation between fault types and, as a result, lowers the overall rate of misclassification. It should be noted that the experimental

Table 6

Test accuracy for fault diagnosis on GPVS.

| Fault          | F1L   | F2L   | F3L   | F4L   | F5L   | F6L  | F7L   | Average |
|----------------|-------|-------|-------|-------|-------|------|-------|---------|
| CNN            | 100   | 64    | 32.33 | 99.67 | 83.33 | 100  | 44.33 | 74.81   |
| LSTM           | 100   | 0     | 23.33 | 100   | 100   | 100  | 77.33 | 71.52   |
| ES-LSTM        | 99.67 | 38.00 | 41.00 | 100   | 100   | 100  | 44.67 | 74.76   |
| KECA-GS-LSTM   | 99.67 | 38.33 | 45.00 | 100   | 100   | 100  | 52.00 | 76.43   |
| DBO-LSTM       | 100   | 41.33 | 35.00 | 100   | 100   | 100  | 60.00 | 76.62   |
| KPCA-QDBO-LSTM | 100   | 34.33 | 62.00 | 100   | 100   | 100  | 59.33 | 79.38   |
| QDBO-LSTM      | 100   | 35.33 | 68.33 | 100   | 100   | 100  | 63.00 | 80.95   |
| BDNPEOLPP      | 83.33 | 95.67 | 100   | 100   | 100   | 1.67 | 100   | 83.38   |
| ES-QLSTM       | 100   | 40.67 | 69.00 | 100   | 100   | 100  | 77.00 | 83.91   |

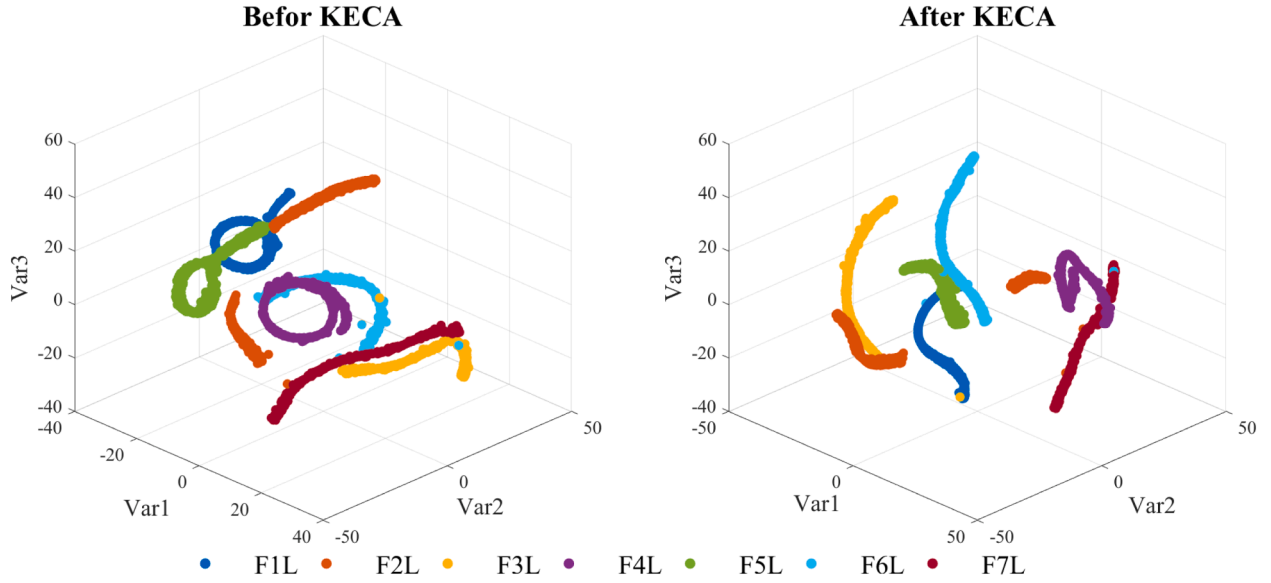


Fig. 10. Visualization graph of the first 3-D features on GPVS.

validation of the ES-QLSTM model mainly focuses on verifying its diagnostic accuracy for complex faults in industrial processes. Consistent with the research focus of the BDNPEOLPP, quantitative analysis of the lightweight computing performance is not included in the present experimental design.

#### 4.2. Grid-connected power generation system

Azzeddine Bakdi developed a case study for GPVS [33], whose flowchart is shown in Fig. 9. Fault data is collected from fault laboratory experiments in photovoltaic (PV) microgrid applications, with each data point comprising 13 dimensions. Scenarios involved PV array faults, inverter faults, grid anomalies, feedback sensor faults, and MPPT controller faults.

This study employs seven out of fourteen distinct fault scenarios, with fault descriptions detailed in Table 5. During the GPVS simulation process, 1000 samples are collected for each fault type, comprising 700 training data points and 300 test data points.

Table 6 shows that the ES-QLSTM achieves an average accuracy of 83.91% in diagnosing seven types of GPVS faults, significantly outperforming the CNN with 74.81%, and the LSTM with 71.52%. Fig. 9 presents the three-dimensional feature vector map. It reveals severe data overlap between F6L and F2L in fault diagnosis, primarily due to the strong coupling between these two faults related to the controller. Traditional models perform poorly for complex faults such as feedback sensor fault F2L, grid anomaly F3L, and boost converter fault F7L. F2L achieves only 2.00% and 0% accuracy under the LSTM, respectively. F3L achieves 29.33% and 23.33% accuracy under the LSTM, respectively. The sensor noise interference of F2L and the dynamic coupling of grid signals of F3L form the main diagnosis difficulties. Entropy space feature extraction suppresses noise and retains the core fault information, providing purer input features for the LSTM. Quantum behavior search endows the QDBO with stronger global optimization capability, and the obtained parameter set enhances the LSTM ability to learn weak fault signals and dynamic coupling features, leading to better performance in diagnosing these two complex faults. The 40.67% accuracy results of F2L from sensor noise, feature overlap with F6L, and insufficient noise suppression and coupled fault differentiation. while F7L achieved 44.33% under the CNN. The ES-QLSTM model improves the accuracies of F2L, F3L, and F7L to 40.67%, 69.00%, and 77.00%, respectively.

Among the improved models, the ES-LSTM achieves an average accuracy of 74.76%, the DBO-LSTM reaches 76.62%, and the QDBO-LSTM attains 80.95%, all lower than the ES-QLSTM. Particularly for F7L faults, the DBO-LSTM accuracy is only 60.00%, while the ES-QLSTM improves the accuracy by 17 %age points, highlighting the effectiveness of its parameter optimization and feature extraction. Meanwhile, the BDNPEOLPP obtains an average accuracy of 83.38% on the GPVS dataset, slightly lower than the ES-QLSTM with 83.91%. While the BDNPEOLPP excels in F2L with 95.67% and F3L with 100% faults due to its powerful dimensionality reduction capability, the ES-QLSTM shows overwhelming advantages in F1L 100% versus with 83.33% and F6L 100% versus with 1.67% faults, verifying the effectiveness of our synergistic framework of entropy space feature extraction and quantum behaved optimization.

From the fault diagnosis confusion matrix in Fig. 11, it can be observed that traditional models such as the SVM, the CNN, and the LSTM exhibit significant misclassification of samples for faults F2L, F3L, and F7L. The non-diagonal elements in the rows corresponding to these faults within the confusion matrix are numerous, indicating blurred classification boundaries. In contrast, the confusion matrix of the ES-QLSTM demonstrates superior classification performance, with no misclassifications for faults F1L, F4L, F5L, and F6L. The number of misclassified samples for F2L, F3L, and F7L is significantly lower than that of traditional models and some modified models. Compared to the DBO-LSTM and the QDBO-LSTM, the ES-QLSTM exhibits a higher proportion of diagonal elements in the confusion matrix rows corresponding to F2L, F3L, and F7L, with fewer samples misclassified into other fault categories. This indicates its ability to more accurately identify complex faults in GPVS and deliver more stable and reliable classification performance.

#### 5. Conclusion

This paper proposes a novel fault diagnosis method named ES-QLSTM. The proposed method uses the kernel entropy component analysis based entropy space feature extraction to process industrial data in entropy space with Rényi entropy as the feature selection metric, which reduces data dimension while preserving high-order fault information and improves feature extraction for high-dimensional nonlinear data. Then, the QDBO algorithm dynamically updates the quantum

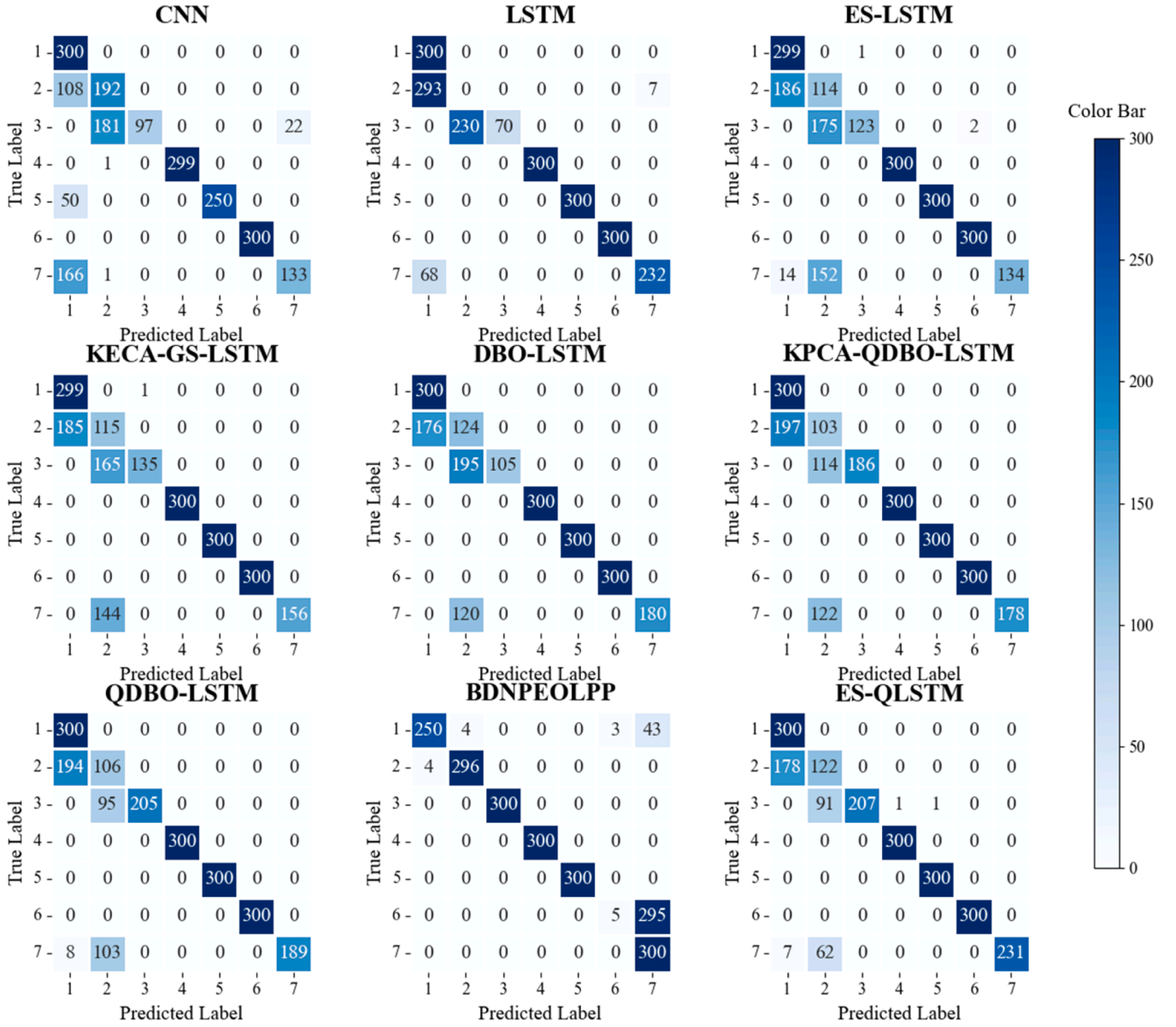


Fig. 11. Fault diagnosis on GPVS.

stride and applies quantum perturbations to optimize LSTM hyperparameters efficiently, thus strengthening the model generalization ability. Experimental results on two industrial benchmarks demonstrate the superiority of the ES-QLSTM. On the TEP dataset, the ES-QLSTM achieves 93.70% average accuracy, which is notably higher than the SVM, the CNN and the single LSTM. On the GPVS dataset, the ES-QLSTM reaches 83.91% average accuracy and also outperforms these comparison methods. Especially for the strongly coupled fault IDV10 in the TEP, the ES-QLSTM obtains 97.50% accuracy, which verifies its strong diagnostic performance.

Future research will incorporate the SHAP analysis to quantify feature contributions and clarify model decision logic, and further develop lightweight model architectures to enhance computational efficiency for real-time industrial applications.

## Nomenclatures

| Symbol                                                 | Symbol Description                                                                                              |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| $X = (x_1, x_2, \dots, x_n)$                           | Raw industrial process data matrix ( $m \times n$ , $m$ is the number of variables, $n$ is the sample size)     |
| $\phi(\cdot)$                                          | Nonlinear mapping function for kernel feature space projection                                                  |
| $K$                                                    | Kernel matrix constructed by Gaussian kernel function ( $n \times n$ )                                          |
| $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$   | Eigenvalue matrix of the centered kernel matrix                                                                 |
| $E = (e_1, \dots, e_n)$                                | Eigenvector matrix of the centered kernel matrix                                                                |
| $\psi_i$                                               | Rényi entropy contribution of the $i^{\text{th}}$ principal component                                           |
| $T_k = [t_1, t_2, \dots, t_k]$                         | Score matrix of dimensionality-reduced features (top $k$ principal components selected by entropy contribution) |
| $\Lambda_k = \text{diag}(\lambda_1, \dots, \lambda_k)$ | Eigenvalue matrix corresponding to top $k$ principal components                                                 |
| $E_k = [e_1, \dots, e_k]$                              | Eigenvector matrix corresponding to top $k$ principal components                                                |
| $P_{\text{train}}$                                     | Preprocessed feature matrix for LSTM input ( $t \times n$ , $t$ is the feature dimension after normalization)   |

(continued on next page)



(continued)

|                                                    |                                                                                                         |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| $\alpha$                                           | Initial learning rate of LSTM (hyperparameter, $\alpha \in [\alpha_{\min}, \alpha_{\max}]$ )            |
| $h$                                                | Number of hidden layer neurons in LSTM (hyperparameter, $h \in [h_{\min}, h_{\max}]$ )                  |
| $\lambda$                                          | L2 regularization coefficient of LSTM (hyperparameter, $\lambda \in [\lambda_{\min}, \lambda_{\max}]$ ) |
| $\theta = (\alpha, h, \lambda)$                    | Hyperparameter vector of LSTM optimized by QDBO                                                         |
| $F(\theta)$                                        | Fitness function of QDBO (classification error rate of LSTM on training set)                            |
| $N$                                                | Population size of QDBO algorithm                                                                       |
| $lb = [\alpha_{\min}, h_{\min}, \lambda_{\min}]^T$ | Lower bound vector of hyperparameter search space                                                       |
| $ub = [\alpha_{\max}, h_{\max}, \lambda_{\max}]^T$ | Upper bound vector of hyperparameter search space                                                       |
| $\theta_{\text{best}}^{(t)}$                       | Optimal hyperparameter solution of QDBO at iteration $t$                                                |
| $\theta_{\text{worst}}^{(t-1)}$                    | Worst hyperparameter solution of QDBO at iteration $t - 1$                                              |
| $T_{\text{max}}$                                   | Maximum number of iterations of QDBO algorithm                                                          |
| $L = 0.1 \cdot (1 - t/T_{\text{max}})$             | Feature length decaying with iterations (for quantum behavior perturbation)                             |
| $u, r$                                             | Uniform random vectors in $[0, 1]$ (for quantum behavior perturbation)                                  |
| $a \in \{-1, 1\}$                                  | Random direction coefficient (for ball-rolling behavior update in QDBO)                                 |
| $b \in [0, 1]$                                     | Random step size coefficient (for ball-rolling behavior update in QDBO)                                 |
| $r_1, r_2$                                         | Random vectors (for foraging behavior update in QDBO)                                                   |
| $X_{\text{new}1}, X_{\text{new}2}$                 | Randomly selected individual positions in QDBO population                                               |
| $H_t$                                              | Hidden state output of LSTM at time step $t$                                                            |
| $C_t$                                              | Cell state of LSTM at time step $t$                                                                     |
| $f_t$                                              | Output of LSTM forget gate at time step $t$                                                             |
| $i_t$                                              | Output of LSTM input gate at time step $t$                                                              |
| $\tilde{C}_t$                                      | Candidate cell state of LSTM at time step $t$                                                           |
| $O_t$                                              | Output of LSTM output gate at time step $t$                                                             |
| $U_f, V_f, \delta_f$                               | Weight matrices and bias vector of LSTM forget gate                                                     |
| $c_i$                                              | True fault category of the $i$ -th sample                                                               |
| $\hat{c}_i = f(p_i; \theta)$                       | Predicted fault category of the $i$ -th sample (LSTM output)                                            |
| $I(\cdot)$                                         | Indicator function (1 if input condition is true, 0 otherwise)                                          |

## CRedit authorship contribution statement

**Yongming Han:** Writing – review & editing, Visualization, Resources, Project administration, Funding acquisition. **Tao Li:** Writing – review & editing, Validation, Resources. **Haorui Liu:** Writing – original draft, Visualization, Methodology, Data curation. **Bo Ma:** Writing – review & editing, Project administration, Funding acquisition. **Shuai Ao:** Writing – original draft, Visualization, Resources, Methodology.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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